

Fig. 8: ESP-01 programmer used for flashing

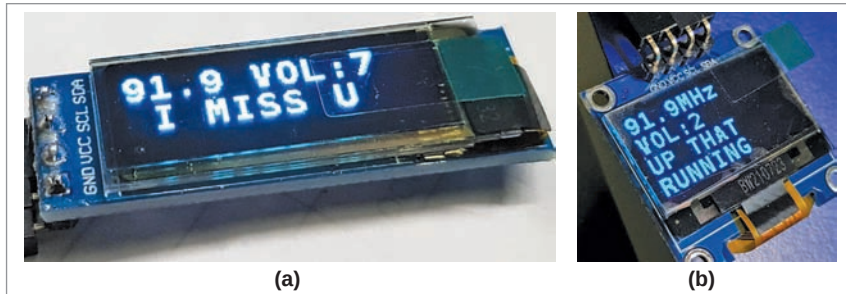


Fig. 9: (a) OLED with 128x32, and (b) OLED with 128x64

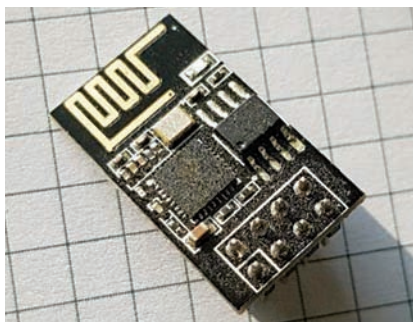


Fig. 10: ESP 01 board

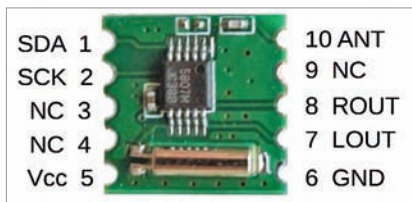


Fig. 11: FM IC RDA5807M IC

Construction and testing

After uploading the source code, connect the setup either on a breadboard or a PCB. The connections for the ESP-01 (U1) with other components such as the OLED, IR module, etc (refer to Fig. 2) are as follows:

1. Connect IO0 to SDA, which is used to control the I2C bus data line. For added safety and reliability, consider adding a 10k-ohm pull-up

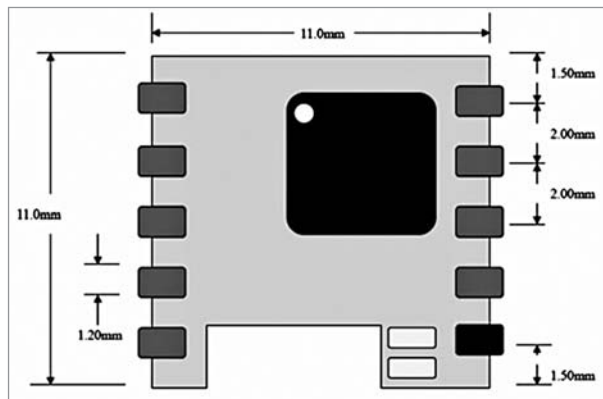


Fig. 12: FM IC RDA5807M IC footprint

resistor to 3.3V.

2. Connect IO2 to SCL, which is used to control the I2C bus clock line. Similar to IO0, you can add a 10k-ohm pull-up resistor to 3.3V for enhanced performance.

3. Connect RX to IR Rx I/P, which is used to receive the IR remote control receiver signal. You can directly connect the IR receiver without any additional components, but ensure it operates at 3.3V.

4. Connect RST to RESET. Pull up the RST pin of the ESP-01 using a 10k-ohm resistor with 3.3V.

5. Connect EN to ENABLE. This pin should be connected to 3.3V via a 12k-ohm resistor.

6. Connect 3.3V to 3.3V and GND to GND on both modules.

The FM module used in this radio is the widely available RDA5807M FM stereo module RRD-102 V2, shown in Fig. 11. Fig. 12 displays the footprint of the FM IC RDA5807M. This module measures 11x11mm and offers 10 user-accessible pins with a 2mm lead pitch.

The connection to the FM IC RDA5807M is straightforward, requiring only seven connections. Pin 1 connects to the SDA pin of the OLED and is also connected to the IO0 pin of the ESP-01. Pin 2 connects to the SCL pin of the OLED and is also linked to the IO2 pin of the ESP-01. Pin 5 connects to 3.3V,

and pin 6 is grounded. Pin 7 is used for the left channel and connects to a 3.5mm audio jack via a 220μF capacitor. Similarly, pin 8 is used for the right channel and connects to a 3.5mm audio jack via a 220μF capacitor. Pin 10, labelled ANT, connects to a 75cm wire, functioning as

an antenna for the FM RF input.

The FM module's size and pad spacing for board layout design are illustrated in Fig. 13. The module measures 11x11mm and features a 2mm pad-to-pad spacing. Additional dimensions are provided in the same figure.

You can use a 2mm header alongside the RDA5807M module, as shown in Fig. 13, or opt for an MSOP to DIP adaptor board or a ZIF socket for quick testing and evaluation.

The FM module is capable of driving earphones or headphones with a 32-ohm load. You can use either 32-ohm headphones, 32-ohm